Frances Tran

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EDUCATION

The University of Texas at Austin *Austin, TX*

B.S. Computational Engineering & B.S. Mathematics | GPA: 3.77/4.00 *Expected May 2028*

Relevant Coursework: Scientific Computation, Engineering Computation, Differential Equations with Linear Algebra, Probability I, Discrete Mathematics, Statics, Dynamics

WORK EXPERIENCE

Product Development Engineering Intern, Bambai Security *06/2026 – 08/2026*

Barcelona, Spain

- Designed a multi-sensor state-machine algorithm processing dynamic inputs (noise, motion, vibration, acceleration) to optimize decision accuracy and minimize false-positive rates in a real-time security alarm system.
- Trained and deployed machine learning models via Edge Impulse onto an ESP32 microcontroller, analyzing high-frequency accelerometer and decibel time-series data to detect events with high confidence.
- Collaborated with the hardware team to integrate and test decision logic on embedded hardware using the Arduino IDE.

Electrical Engineering Intern, DELL Technologies *02/2024 – 04/2024*

Austin, TX

- Demoed a working robot-dog mobility-assistance device to a panel of Dell engineers as part of a 7-person team, placing 2nd out of all intern teams given the same challenge.
- Implemented Python motor-control logic on a Raspberry Pi and verified voltage/power calculations using bench-level DC power supplies, multi-meters, and MATLAB, validating a 4-hour battery runtime target.
- Designed system block diagrams to communicate interconnecting hardware/software elements to cross-functional stakeholders.

Summer Coding Camp Instructor, DOit Coding *06/2024 – 08/2024*

Austin, TX

- Led 3-hour Python and Scratch (MIT) coding sessions 5 days/week for students ages 7–11, achieving a 100% project-completion rate across all camp sessions.

PROJECTS

Content-Sensitivity Text Classifier (DistilBERT) *08/2025 – Present*

- Fine-tuned a DistilBERT transformer model (Hugging Face) on a labeled dataset to classify short-form text by content sensitivity, achieving 87% accuracy on held-out test data.
- Evaluated model robustness on a self-authored set of adversarial phrases to surface edge cases, informing potential applicability to real-time content-moderation pipelines.

GARCH Volatility Forecasting & Mean-Reversion Strategy *Texas UCF*

- Implemented a GARCH-based volatility-forecasting model, adapted from academic literature, to trigger a mean-reversion trading strategy above a defined volatility threshold.
- Backtested cumulative returns across three tickers (AAPL, RH, QCOM); presented mixed but rigorously evaluated results and strategy rationale to Texas UCF.

ACTIVITIES

- Texas Undergraduate Computational Finance (Texas UCF) — 10/2025–present
- Engineering Study Abroad, Barcelona, Spain — 05/2026–08/2026
- Engineering Study Abroad, Copenhagen, Denmark — 05/2025–06/2025
- Texas Run Club — 09/2024–present

SKILLS

Languages/Tools: Python, C++, MATLAB, R, SQL (basic), SolidWorks (CAD), Tableau

Platforms: Edge Impulse, Arduino IDE, Hugging Face, GitHub